

# Data Visualization with Power BI

## Dataset Guide for Students

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Pick one of the four datasets below. Each one has a description, a list of what every column means, and five questions you can answer using Power BI. Every question includes the chart type to use and the steps to build it.

You do not need to answer all five. Pick two or three that look interesting and build them well.

All four files have already been cleaned for you. There is nothing to fix before you start. Load the file and go straight to building charts.

### Which dataset should I pick?

| Dataset              | Good for                                                                                               | Difficulty |
|----------------------|--------------------------------------------------------------------------------------------------------|------------|
| Superstore Sales     | Easiest. Best if you want a dashboard that looks like something a real business would use. Great maps. | Easiest    |
| Video Game Sales     | Easy. Best if you like games and want to compare genres, consoles, and countries.                      | Easy       |
| NFL Team Stats       | Medium. Best if you follow football and want to compare teams and seasons.                             | Medium     |
| S&P 500 Stock Prices | Hardest. Best if you want to work with dates and line charts. It is a big file, so it loads slower.    | Hardest    |

### Getting your data into Power BI

- Open Power BI Desktop. On the Home ribbon click Get Data, then Text/CSV.
- Find your CSV file, click Open, then click Load. Do not click Transform Data unless you are told to.
- Your columns now appear in the Data pane on the right side of the screen.
- Click a chart type in the Visualizations pane first, then drag column names into the boxes underneath it.

### Four things that trip everyone up the first time

- Power BI adds up numbers by default. If a chart shows a giant number that makes no sense, click the small arrow next to the field name in the Visualizations pane and switch from Sum to Average, or to Count.
- Year columns are not really numbers. Power BI will try to add years together and give you something like 4,032,144. Click the Year column in the Data pane, go to the Column tools ribbon, and set Summarization to Don't summarize.
- To show only the top 10 of something, drag your category into the Filters pane, set Filter type to Top N, enter 10, and drag the number you care about into the By value box.
- To sort a bar chart, hover over the chart, click the three dots in the corner, choose Sort axis, and pick the field to sort by.

## Dataset 1: Superstore Sales

**File:** Sample\_-\_Superstore.csv

**Size:** 9,994 rows, 21 columns

**Difficulty:** Easiest

This is four years of orders from a made-up office supply store in the United States, covering January 2014 through December 2017. Every row is one product on one order, so a single order that included a chair and a box of pens shows up as two rows. The store has three big product categories (Furniture, Office Supplies, and Technology), sells in 49 states, and serves three types of customers.

What makes this dataset interesting is that it tracks both Sales (money coming in) and Profit (money actually kept). Those two do not always agree, and finding where they disagree is the whole point.

### Column descriptions

| Column        | What it means                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| Row ID        | A number for each row. Ignore it.                                                                                   |
| Order ID      | The order this product belonged to. The same ID appears on several rows when someone bought several things at once. |
| Order Date    | The day the customer placed the order.                                                                              |
| Ship Date     | The day the order left the warehouse.                                                                               |
| Ship Mode     | Shipping speed: Standard Class, Second Class, First Class, or Same Day.                                             |
| Customer ID   | A code for the customer. There are 793 different customers.                                                         |
| Customer Name | The customer's name.                                                                                                |
| Segment       | Type of customer: Consumer, Corporate, or Home Office.                                                              |
| Country       | Always United States, so this column is not useful.                                                                 |
| City          | City the order shipped to. There are 531 of them.                                                                   |
| State         | State the order shipped to. This one works great on a map.                                                          |
| Postal Code   | ZIP code. Power BI will try to add these up, so set it to Don't summarize.                                          |
| Region        | Central, East, South, or West.                                                                                      |
| Product ID    | A code for the product.                                                                                             |
| Category      | Furniture, Office Supplies, or Technology.                                                                          |
| Sub-Category  | A narrower group inside the category, such as Chairs, Phones, Binders, Paper, or Copiers. There are 17.             |
| Product Name  | The full name of the product.                                                                                       |
| Sales         | How much money the customer paid for this line, in dollars.                                                         |
| Quantity      | How many units they bought.                                                                                         |
| Discount      | The discount given, written as a decimal. 0.2 means 20 percent off, 0 means full price.                             |

| Column | What it means                                                                                                  |
|--------|----------------------------------------------------------------------------------------------------------------|
| Profit | How much the store actually made on this line, in dollars. This can be negative, meaning the store lost money. |

## Questions to answer

### 1. Which product sub-category brings in the most sales, and which one actually loses the store money?

**Chart:** Two bar charts side by side.

**How to build it:** Build a Clustered bar chart with Sub-Category on the Y axis and Sales in the Values box. Then build a second one exactly the same but with Profit instead of Sales. Sort both from largest to smallest and compare. Watch for a bar that points the wrong way.

### 2. Do sales go up at certain times of year, like around the holidays?

**Chart:** Line chart.

**How to build it:** Use a Line chart. Put Order Date on the X axis and Sales in the Values box. Power BI will group the dates by year automatically. Click the split-arrow icon at the top of the chart to drill down into months.

### 3. Which states buy the most from the store?

**Chart:** Filled map.

**How to build it:** Use a Filled map. Drag State into the Location box and Sales into the Tooltips box. Darker states sold more. If the map does not appear, ask for help turning on map visuals.

### 4. Which type of customer is the most profitable: Consumer, Corporate, or Home Office?

**Chart:** Donut chart.

**How to build it:** Use a Donut chart. Put Segment in the Legend box and Profit in the Values box.

### 5. Does giving customers a bigger discount hurt the store's profit?

**Chart:** Scatter chart.

**How to build it:** Use a Scatter chart. Put Discount on the X axis, Profit on the Y axis, and Product Name in the Values box so each product gets its own dot. Look at what happens to the dots once the discount passes 0.3.

## Watch out for

- If product names show up with strange symbols in them, click Transform Data and set File Origin to Western European (Windows).
- Country is the same for every row, so it will make a very boring chart.

## Dataset 2: Video Game Sales

**File:** vgsales\_clean.csv

**Size:** 16,323 rows, 11 columns

**Difficulty:** Easy

This is a list of every video game that sold more than 100,000 copies, from 1980 up through about 2016. Each row is one game on one console, which means a game released on both PlayStation and Xbox shows up twice. The numbers are copies sold in millions, split up by part of the world, so 82.74 means about 82.74 million copies.

The most interesting thing in this dataset is that Japan does not buy the same games as the rest of the world.

### Column descriptions

| Column       | What it means                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rank         | The game's place on the all-time global sales list. Rank 1 sold the most. Set this to Don't summarize.                                                |
| Name         | Title of the game.                                                                                                                                    |
| Platform     | The console or system it came out on, such as Wii, PS3, X360, DS, or PC. There are 31.                                                                |
| Year         | Year the game was released, from 1980 to 2016. Set this to Don't summarize.                                                                           |
| Genre        | The kind of game. There are 12: Action, Adventure, Fighting, Misc, Platform, Puzzle, Racing, Role-Playing, Shooter, Simulation, Sports, and Strategy. |
| Publisher    | The company that released the game. There are 576. A few older games are listed as Unknown.                                                           |
| NA_Sales     | Copies sold in North America, in millions.                                                                                                            |
| EU_Sales     | Copies sold in Europe, in millions.                                                                                                                   |
| JP_Sales     | Copies sold in Japan, in millions.                                                                                                                    |
| Other_Sales  | Copies sold everywhere else, in millions.                                                                                                             |
| Global_Sales | Total copies sold worldwide, in millions. This is the other four columns added together.                                                              |

### Questions to answer

#### 1. Which genre of game has sold the most copies worldwide?

**Chart:** Bar chart.

**How to build it:** Use a Clustered bar chart. Put Genre on the Y axis and Global\_Sales in the Values box. Sort it from largest to smallest.

#### 2. What are the ten best selling video games of all time?

**Chart:** Bar chart with a Top 10 filter.

**How to build it:** Clustered bar chart with Name on the Y axis and Global\_Sales in the Values box. Then drag Name into the Filters pane, set Filter type to Top N, enter 10, and drag Global\_Sales into the By value box.

### 3. Do people in Japan buy different kinds of games than people in North America?

**Chart:** Stacked bar chart, or two bar charts side by side.

**How to build it:** Use a Stacked bar chart with Genre on the Y axis and both NA\_Sales and JP\_Sales in the Values box. Role-Playing games are the ones to look at.

### 4. How did total video game sales change from 1995 to 2015? Did the industry keep growing?

**Chart:** Line chart.

**How to build it:** Line chart with Year on the X axis and Global\_Sales in the Values box. Drag Year into the Filters pane and slide the range to 1995 through 2015, so the thin early years do not squash the interesting part.

### 5. Which console sold the most games, and which publisher sold the most copies?

**Chart:** Bar chart, or a treemap.

**How to build it:** Clustered bar chart with Platform on the Y axis and Global\_Sales in the Values box. Then swap Platform for Publisher and add a Top 10 filter, since 578 publishers will not fit on one chart. A Treemap looks good for this one too.

### Watch out for

- Sales numbers are copies sold, not dollars. A value of 82.74 means about 82.74 million copies.
- A game released on more than one console appears once for each console, so do not be surprised to see the same title twice.

## Dataset 3: NFL Team Stats

**File:** team\_stats\_2003\_2023.csv

**Size:** 672 rows, 35 columns

**Difficulty:** Medium

This is one row for every NFL team in every regular season from 2003 through 2023, which works out to 32 teams times 21 seasons. It covers how many games they won, how many points they scored and gave up, how many yards they gained running and passing, how many touchdowns they scored, how many times they turned the ball over, and how many penalties they took.

Because this is season totals and not individual games, it is small and easy to work with. Note that seasons through 2020 had 16 games and seasons from 2021 on had 17, so totals from different eras are not perfectly fair to compare. Use averages per game if you want to be careful about it.

### Column descriptions

| Column               | What it means                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------|
| year                 | The season. Set this to Don't summarize.                                                              |
| team                 | Full team name.                                                                                       |
| wins                 | Games won that season.                                                                                |
| losses               | Games lost that season.                                                                               |
| win_loss_perc        | Win percentage, from 0 to 1. A 0.875 means the team won 87.5 percent of its games.                    |
| points               | Total points the team scored.                                                                         |
| points_opp           | Total points scored against them.                                                                     |
| points_diff          | Points scored minus points allowed. Positive is good.                                                 |
| mov                  | Margin of victory per game. This column is blank for about half the rows, so use points_diff instead. |
| g                    | Games played. 16 for older seasons, 17 from 2021 on.                                                  |
| total_yards          | Total offensive yards gained.                                                                         |
| plays_offense        | Number of offensive plays run.                                                                        |
| yds_per_play_offense | Average yards gained per play.                                                                        |
| turnovers            | Times the team lost the ball, by fumble or interception.                                              |
| fumbles_lost         | Fumbles the team lost to the other team.                                                              |
| first_down           | Total first downs earned.                                                                             |
| pass_cmp             | Completed passes.                                                                                     |
| pass_att             | Passes attempted.                                                                                     |
| pass_yds             | Yards gained through the air.                                                                         |
| pass_td              | Touchdowns scored by passing.                                                                         |

| Column               | What it means                                        |
|----------------------|------------------------------------------------------|
| pass_int             | Interceptions thrown.                                |
| pass_net_yds_per_att | Average yards per pass attempt.                      |
| pass_fd              | First downs earned by passing.                       |
| rush_att             | Running plays attempted.                             |
| rush_yds             | Yards gained on the ground.                          |
| rush_td              | Touchdowns scored by running.                        |
| rush_yds_per_att     | Average yards per running play.                      |
| rush_fd              | First downs earned by running.                       |
| penalties            | Number of penalties called on the team.              |
| penalties_yds        | Yards lost to penalties.                             |
| pen_fd               | First downs the team gave the other team by penalty. |
| score_pct            | Percent of drives that ended in a score.             |
| turnover_pct         | Percent of drives that ended in a turnover.          |
| exp_pts_tot          | An advanced statistic. You can ignore this column.   |
| ties                 | Number of tied games. Blank means zero ties.         |

## Questions to answer

### 1. Which team scored the most touchdowns on average per season, and did they get them by running or by passing?

**Chart:** Stacked bar chart.

**How to build it:** Use a Stacked bar chart. Put team on the Y axis, then put both pass\_td and rush\_td in the Values box. Change both of them from Sum to Average using the arrow next to the field name. The full bar is total touchdowns and the two colors show how they got there.

### 2. Which team has the best win percentage across all 21 seasons?

**Chart:** Bar chart.

**How to build it:** Clustered bar chart with team on the Y axis and win\_loss\_perc in the Values box, changed from Sum to Average. Sort it largest to smallest.

### 3. Do teams that score more points actually win more games?

**Chart:** Scatter chart.

**How to build it:** Scatter chart with points on the X axis, wins on the Y axis, and team in the Values box. Add year to the Filters pane and pick one season, otherwise every team appears 21 times and the chart gets crowded.

### 4. Has the NFL become more of a passing league since 2003?

**Chart:** Line chart with two lines.

**How to build it:** Line chart with year on the X axis and both pass\_yds and rush\_yds in the Values box, both changed from Sum to Average. Watch how the gap between the two lines changes.

### 5. Pick your favorite team. How have they done over the last 21 seasons?

**Chart:** Line chart plus a slicer.

**How to build it:** Line chart with year on the X axis and wins in the Values box. Then add a Slicer visual, drag team into it, and click your team. The slicer will filter every chart on the page at once, which is the whole reason Power BI is useful.

### Watch out for

- There are 37 team names instead of 32 because some teams moved cities or changed names. The Rams appear as both St. Louis Rams and Los Angeles Rams, the Chargers as both San Diego and Los Angeles, the Raiders as both Oakland and Las Vegas, and Washington appears under three different names.
- The mov and ties columns have a lot of blanks. Skip them.

## Dataset 4: S&P 500 Stock Prices

**File:** all\_stocks\_5yr.csv

**Size:** 619,040 rows, 7 columns

**Difficulty:** Hardest

This is five years of daily stock prices, from February 2013 to February 2018, for all 505 companies in the S&P 500. That index is a list of the 500 biggest publicly traded companies in the United States. Each row is one company on one trading day, which is why the file is so large.

A stock price is what one share of a company costs. When people say a company had a good year, they usually mean this number went up. There are only seven columns, so it is simple to understand, but there is a lot of data and only one thing to group by, which makes it the trickiest of the four to build charts from. Pick it if you want to work with dates and line charts.

### Column descriptions

| Column | What it means                                                                                                                                                                                          |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| date   | The trading day, written as year-month-day. Only weekdays appear, because the stock market is closed on weekends and holidays.                                                                         |
| open   | The price of one share when trading started that morning, in dollars.                                                                                                                                  |
| high   | The highest price the share reached that day.                                                                                                                                                          |
| low    | The lowest price the share reached that day.                                                                                                                                                           |
| close  | The price of one share when trading ended that afternoon. This is the number people usually mean by the stock price.                                                                                   |
| volume | How many shares were bought and sold that day.                                                                                                                                                         |
| Name   | The company's ticker symbol, which is its short code on the stock market. AAPL is Apple, MSFT is Microsoft, NKE is Nike, DIS is Disney, NFLX is Netflix, SBUX is Starbucks, NVDA is NVIDIA, F is Ford. |

### Questions to answer

#### 1. How did Apple's stock price change over these five years?

**Chart:** Line chart.

**How to build it:** Line chart with date on the X axis and close in the Values box, changed from Sum to Average. Then drag Name into the Filters pane and select AAPL. Do this filter step first, or the chart will try to draw all 505 companies at once.

#### 2. Which grew faster over five years, Netflix or Disney?

**Chart:** Line chart with two lines.

**How to build it:** Same line chart as above, but drag Name into the Legend box and use the Filters pane to keep only NFLX and DIS. Each company gets its own colored line. NVDA against F is an even more dramatic pair if you want to try another.

#### 3. Which ten companies had the most shares traded per day on average?

**Chart:** Bar chart with a Top 10 filter.

**How to build it:** Clustered bar chart with Name on the Y axis and volume in the Values box, changed from Sum to Average. Then drag Name into the Filters pane, set Filter type to Top N, enter 10, and use average of volume as the By value. The answer says something about which companies people trade most often, which is not the same as which are biggest.

#### 4. Which company had the highest share price of anyone in the whole dataset?

**Chart:** Table, or a bar chart with a Top 10 filter.

**How to build it:** Use a Table visual with Name and close. Click the arrow next to close and change Sum to Maximum, then sort that column largest to smallest. The winner is a company most people have never heard of, and the runner-up is one everybody knows.

#### 5. Were there any days when the whole market traded unusually heavily?

**Chart:** Line chart.

**How to build it:** Line chart with date on the X axis and volume in the Values box, left as Sum so it adds up all 505 companies. Big spikes are days when something happened in the news. Look at early February 2018.

#### Watch out for

- This file has over 600,000 rows. It will take a minute to load and charts will redraw a little slowly. That is normal, not a broken computer.
- Always filter down to a few companies before making a line chart. A chart with 505 lines on it is unreadable.
- You have to know the ticker symbol to find a company. A few worth knowing: AAPL Apple, AMZN Amazon, MSFT Microsoft, GOOGL Google, NFLX Netflix, DIS Disney, NKE Nike, SBUX Starbucks, MCD McDonald's, WMT Walmart, NVDA NVIDIA, EA Electronic Arts, F Ford, BA Boeing, KO Coca-Cola.
- A few rows are missing an open, high, or low value. There are only about 27 of them out of 619,040, so they will not affect anything.
- A high share price does not mean a company is bigger or better, only that it split its shares fewer times.

## Quick reference: which chart do I use?

A chart is only good if it answers the question you asked. Here is the short version.

| Chart type          | Use it when you are                                           | Example from these datasets                        |
|---------------------|---------------------------------------------------------------|----------------------------------------------------|
| Bar or column chart | Comparing a number across categories                          | Sales by product category, wins by team            |
| Line chart          | Showing how something changes over time                       | Stock price by date, game sales by year            |
| Stacked bar chart   | Comparing categories while also showing what they are made of | Touchdowns by team, split into passing and rushing |
| Scatter chart       | Checking whether two numbers are related                      | Discount versus profit, points scored versus wins  |
| Pie or donut chart  | Showing parts of a whole, when there are only a few parts     | Profit split across three customer segments        |
| Map                 | Anything with states, countries, or cities                    | Sales by state                                     |
| Card                | One single important number, shown big                        | Total sales, total profit                          |
| Table               | When the exact numbers matter more than the shape             | Top 10 list with precise values                    |

### Two rules worth remembering

- Pie charts stop working past about five slices. If you have 17 sub-categories, use a bar chart.
- If your chart needs a paragraph of explanation before anyone understands it, it is the wrong chart.